



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT 2

AIM:

To use var, let, and const, and demonstrate ES6 features like template literals and destructuring by building a billing calculator.

Software/Tools Required:

- Visual Studio Code (VS Code)
- Web Browser (Google Chrome/Microsoft Edge)
- HTML5
- JavaScript (ES6)

Theory:

JavaScript ES6 (ECMAScript 2015) introduced several modern features that make programming more efficient, readable, and easier to maintain. Among these features are var, let, and const, which are used for variable declaration. The var keyword has function scope and allows variables to be redeclared and updated. The let keyword provides block scope and allows variables to be updated but not redeclared within the same scope. The const keyword is also block-scoped and is used to declare constants whose values cannot be reassigned after initialization.

Template literals are another important ES6 feature that simplify string creation and formatting. They use backticks (```) instead of quotation marks and allow variables and expressions to be embedded directly into strings using `${expression}`. This eliminates the need for string concatenation and improves code readability.

Destructuring is a convenient JavaScript feature that enables the extraction of values from arrays or properties from objects into separate variables. It reduces repetitive code, makes variable assignment more concise, and improves code clarity, especially when working with complex data structures.



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In this experiment, a billing calculator is developed using HTML and JavaScript. The application accepts user inputs such as product details, quantity, and price, and performs calculations to determine the total bill. Variables are declared using var, let, and const wherever appropriate, while template literals are used to display formatted billing information. Destructuring is applied to efficiently access and manage object or array values. This experiment demonstrates the practical implementation of modern JavaScript features in developing a simple, interactive, and user-friendly web application.

Output:

Result:

Successfully implemented a billing calculator using JavaScript ES6 features, including var, let, const, template literals, and destructuring, to calculate and display billing details accurately.



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CASE STUDY 2

AIM:

Develop the webpage for an online shopping website with using html,css and html tags and calculate bill,customer name,mobile no,invoice no,product no,quantity(kg),rate per kg,discount,GST,packing charges,payment mode(cash/card/UPI), Membership(yes/no).

PROGRAM CODE:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Soumya Shopping Cart</title>
7   <style>
8     body {
9       font-family: Arial, sans-serif;
10      background: #f4f8ff;
11      margin: 0;
12      padding: 20px;
13    }
14
15    .container {
16      max-width: 1000px;
17      margin: auto;
18      background: white;
19      padding: 25px;
20      border-radius: 12px;
21      box-shadow: 0 4px 12px rgba(0,0,0,0.08);
22    }
23
24    h1 {
25      text-align: center;
26      color: #2c5aa0;
27    }
```



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```
119 </head>
120 <body>
121   <div class="container">
122     <h1>Soumya Shopping Cart</h1>
123
124     <div class="grid">
125       <div class="field">
126         <label>Customer Name</label>
127         <input type="text" id="customerName">
128       </div>
129
130       <div class="field">
131         <label>Mobile No</label>
132         <input type="text" id="mobileNo">
133       </div>
134
135       <div class="field">
136         <label>Invoice No</label>
137         <input type="text" id="invoiceNo">
```



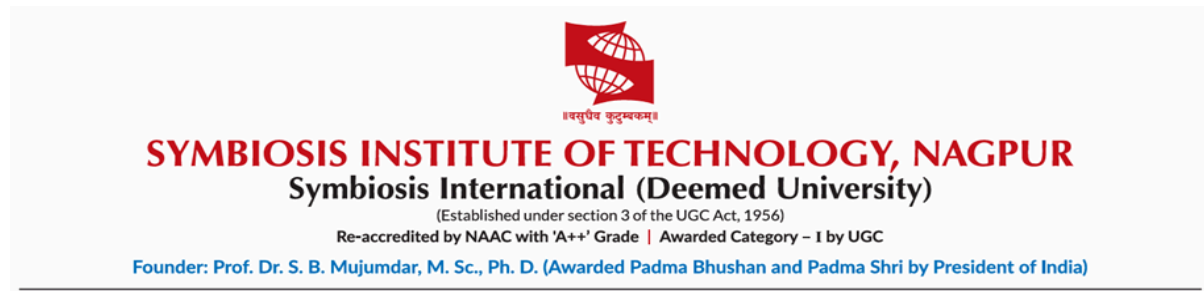
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```
134
135     <div class="field">
136         <label>Invoice No</label>
137         <input type="text" id="invoiceNo">
138     </div>
139
140     <div class="field">
141         <label>Packing Charges</label>
142         <input type="number" id="packing" min="0" value="0">
143     </div>
144
145     <div class="field">
146         <label>Payment Mode</label>
147         <select id="paymentMode">
148             <option>Cash</option>
149             <option>Card</option>
150             <option>UPI</option>
151         </select>
152     </div>
153
154     <div class="field">
155         <label>Membership</label>
156         <select id="membership">
```



OUTPUT:

The screenshot shows a web browser window with the title 'Soumya Shopping Cart'. The URL in the address bar is 'C:/Users/SOUMYA%20MEHAR/OneDrive/Desktop/Projects/JS%20Lab/Case%20study%202/index.html'. The web page has a light blue background. The main content area is a white box with the title 'Soumya Shopping Cart' in blue. It contains several input fields for customer details: 'Customer Name' (filled with 'Soumya'), 'Mobile No' (filled with '9588446174'), 'Invoice No' (filled with '24070521200'), 'Packing Charges' (filled with '2'), 'Payment Mode' (a dropdown menu with 'Cash' selected), and 'Membership' (a dropdown menu with 'Yes' selected). Below these fields is a table with columns: 'Product Name', 'Quantity (kg)', 'Rate per kg', 'Discount (%)', and 'GST (%)'. The table has one row with 'Sugar', '5', '10', '20', and '0.18'. To the right of the table is a red 'Remove' button. Below the table are two buttons: 'Add Item' and 'Calculate Bill'. At the bottom of the white box is a summary table with the following data: Customer Name: Soumya, Mobile No: 9588446174, Invoice No: 24070521200, Packing Charges: 2.00, Payment Mode: Cash, Membership: Yes, Sugar (5 kg): 40.07, and Total Bill: 42.07. The browser's taskbar at the bottom shows various icons and the system clock indicating 08:19 PM on 05-08-2026.

RESULT:

The online shopping billing webpage was successfully developed using HTML, CSS, and JavaScript. The program accepted customer and product details, calculated the total bill including discount, GST, and packing charges, and displayed the final invoice correctly. It demonstrated the use of user input, arithmetic calculations, objects, template literals, and object destructuring in JavaScript.